LifeLedger Consortium Platform

Comprehensive User Interface Specification, Button Directory & System Architecture Report

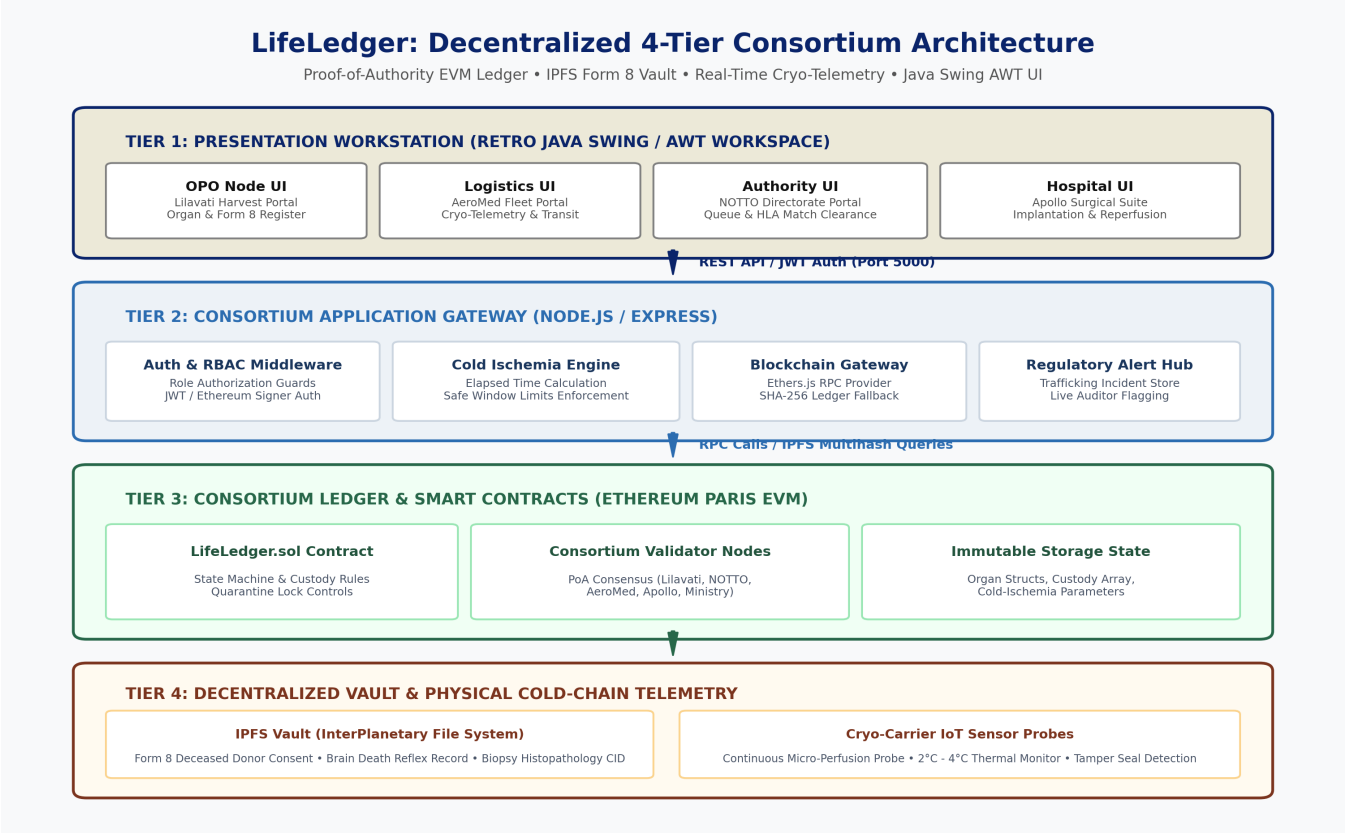
System Identity:	LifeLedger (PoA EVM Consortium v2.0)	**Date of Release:**	October 25, 2026
Author / Lead:	Soham Bhale (TrueDose Project)	**Compliance:**	NOTA / THOTA & NOTTO Form 8
UI Look & Feel:	Retro Java Swing / AWT Workstation	**Test Verification:**	41/41 Automated Invariants (100%)

1. Executive Summary & Problem Resolution

LifeLedger is a decentralized medical blockchain network designed to address critical vulnerabilities in the human organ transplantation supply chain: black-market commercial trafficking, falsified deceased donor consent papers, and ischemic cellular death caused by cold-chain transit delays. By combining an immutable Proof-of-Authority (PoA) Ethereum Virtual Machine (EVM) ledger with decentralized IPFS document storage, real-time cryo-perfusion telemetry (2°C - 4°C), and an authentic retro Java Swing / AWT desktop GUI, LifeLedger enforces mathematical non-repudiation across hospitals, organ procurement organizations (OPOs), and national health authorities.

2. Consortium System Architecture & Infrastructure

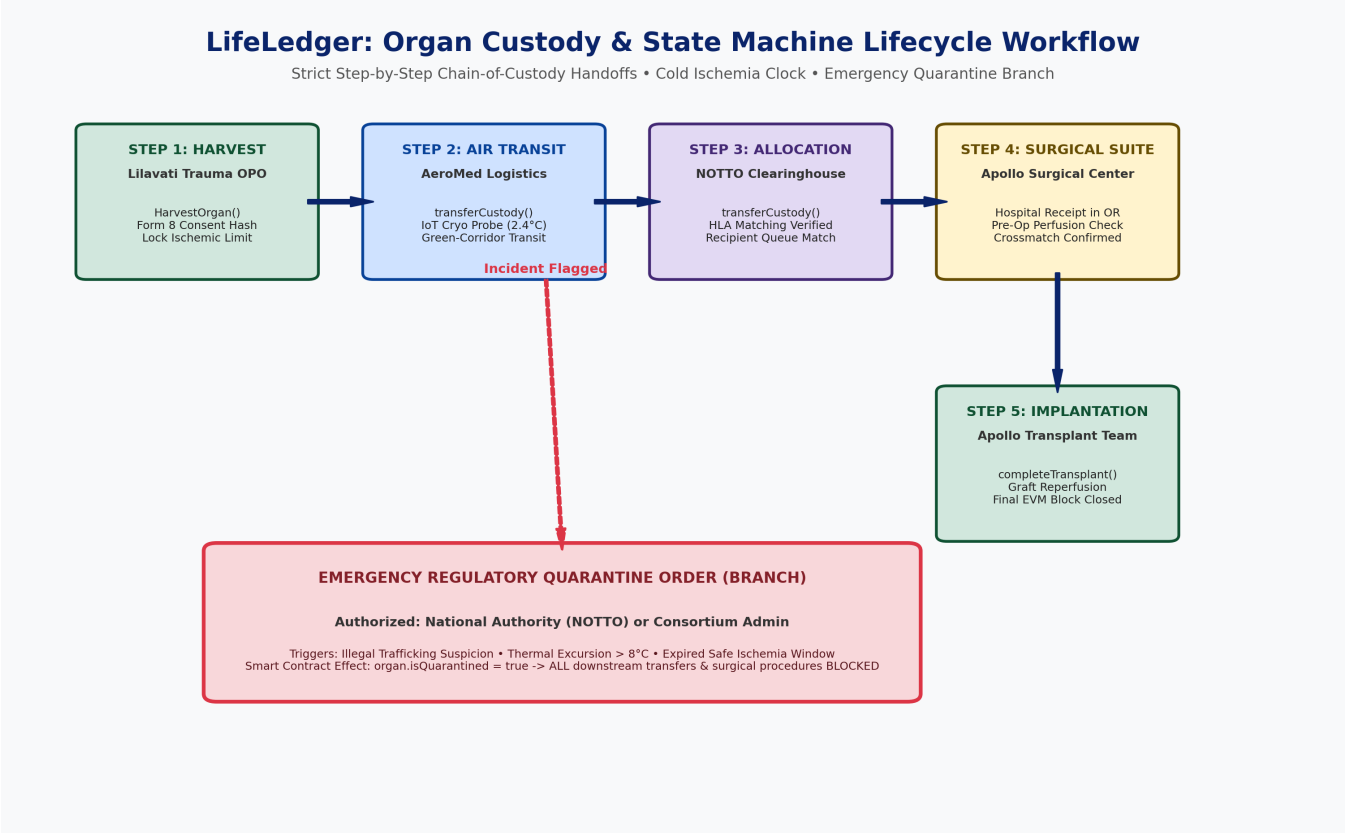
The LifeLedger architecture is divided into 4 decoupled, fault-tolerant tiers ensuring that real-time sensor telemetry, legal documents, and blockchain state transitions synchronize with millisecond finality:



Architectural Tier	Key Technologies	Consortium Responsibility
Tier 1: Workstation UI	React 19, Vite, Tailwind CSS, Retro Java Swing Look & Feel	Workstation menus (JMenuBar), beveled buttons (JToolBar), registry table (JTable), and dialogs (JDialog).
Tier 2: API Gateway	Node.js, Express, Ethers.js, JWT, SHA-256 Engine	RBAC authorization middleware, dynamic cold ischemia countdown engine, and live regulatory alert dispatch.
Tier 3: Consortium EVM	Solidity v0.8.20, Hardhat, Paris EVM, PoA Consensus	Smart contract (LifeLedger.sol) enforcing custody authorization, zero double-allocation, and quarantine locks.
Tier 4: Vault & IoT	IPFS (InterPlanetary File System), Cryo Probes, QR/RFID	Cryptographic storage of Form 8 donor consent papers, brain death reflex evaluations, and continuous 2-4°C sensors.

3. End-to-End Organ Custody & State Machine Lifecycle

Every human organ registered on LifeLedger moves through an immutable state machine where state transitions are strictly governed by the smart contract. Cold-ischemia time windows (Heart: 4h, Liver: 12h, Kidney: 24h) are locked at harvest, and any regulatory entity can execute an instant consortium-wide quarantine freeze.



Lifecycle Phase	Authorized Node	Smart Contract Action & Rules Enforced
1. Deceased Donor Harvest	Lilavati Trauma OPO (OPO_ROLE)	Calls harvestOrgan(). Locks donor ID, ABO blood group, HLA markers, initial cryo temp, and Form 8 IPFS CID.
2. Cold-Chain Air Transit	AeroMed Logistics (LOGISTICS_ROLE)	Calls transferCustody(). Dispatches green-corridor flight, logs probe telemetry (2.4°C), and updates location.
3. Allocation Clearance	NOTTO Directorate (AUTHORITY_ROLE)	Verifies histocompatibility crossmatch, audits recipient prioritization queue, and clears hospital destination.
4. Surgical Suite Handoff	Apollo Surgical Hospital (HOSPITAL_ROLE)	Receives organ in Operating Theater. Surgeon verifies seal integrity and cold-perfusion viability.
5. Implantation Sign-Off	Apollo Surgical Hospital (HOSPITAL_ROLE)	Calls completeTransplant(). Reperfusion verified. Organ state permanently marked as Transplanted.
Emergency Quarantine	Health Authority / Admin (ADMIN_ROLE)	Calls quarantineOrgan(). Triggers immediate consortium-wide freeze. Blocks all further transfers.

4. Comprehensive Button & UI Control Directory

The following exhaustive directory details every interactive button, menu item, dropdown, and dialog control across the entire LifeLedger desktop application, including their technical event triggers and system effects:

4.1 Navigation Bar, JMenuBar & JToolBar Controls

Control Element	Type	Action & Operational Effect
File Menu	JMenu	Opens file dropdown: *Harvest Donor Organ* (opens harvest modal), *Print Specimen Manifest*, and *Exit Workstation*.
View Menu	JMenu	Switches workspace frame: *Organ Custody Registry* (F2), *Provenance Inspector* (F3), and *Consortium Explorer* (F4).
Stakeholders Menu	JMenu	Switches active Ethereum signing identity between NOTTO Admin, Lilavati OPO, AeroMed, NOTTO Allocation, and Apollo.
Help Menu	JMenu	Displays: *System Architecture & Viva Defense Manual* (F1), *API Health Status Check*, and *Sample REST JSON*.
[Harvest Organ]	JToolBar Btn	Available to OPO & Admin. Launches the CreateBatchModal dialog to record a newly procured donor graft.
[Organ Registry]	JToolBar Btn	Switches active view to the main organ inventory grid (DashboardPage) with live cold-ischemia clocks.
[Provenance]	JToolBar Btn	Navigates to the public verification inspection terminal (VerifyPage) for cryptographic donor audits.
[Consortium]	JToolBar Btn	Opens the Proof-of-Authority node explorer (ExplorerPage) displaying validator health and transaction pings.
[Refresh (F5)]	JToolBar Btn	Re-fetches fresh ledger records, updates elapsed cold-ischemia hours, and recalculates active incident cases.
Quick Presets (Kidney / Heart / Liver)	JToolBar Btns	Instantly switches the active workstation node to match the operational team responsible for that organ.
Consortium Role (JComboBox)	JComboBox	Dropdown that dynamically changes the active user context (OPO, Logistics, Authority, Hospital, Admin).
[Defense Guide [F1]]	JToolBar Btn	Opens the Project Defense Guide modal containing 1-minute demo sequence and examiner viva talking points.

4.2 Organ Custody Registry Table (DashboardPage) Actions

Control Element	Type	Action & Operational Effect
Status Filter Tabs	JButton Filter	Filters registry rows: *All Records*, *In Transit (Cryo)*, *Allocation Cleared*, *Transplanted*, or *Quarantined*.
Search Registry Input	JTextField	Live text filter matching Organ ID, Organ Type, Blood Group, Donor ID, or Recipient Hospital.
Table Row Selection	JTable Click	Highlights selected row in classic navy #0A246A and activates contextual custody actions.
[Inspect] (Eye Icon)	Table Action Btn	Navigates directly to VerifyPage with this specific organ pre-loaded for full donor provenance audit.
[Tag] (QrCode Icon)	Table Action Btn	Opens QRCodeModal displaying printable Cryo-Carrier Specimen Tag with QR code and manifest.
[Form 8] (Award Icon)	Table Action Btn	Opens CertificateModal displaying the official Form 8 Deceased Donor Consent and Brain Death Clearance.
[Transfer] (Truck Icon)	Table Action Btn	Enabled only for current custodian. Opens TransferModal to execute blockchain custody handoff.
[Implant] (Heart Icon)	Table Action Btn	Enabled only for Hospital nodes when organ is received. Executes final surgical sign-off and reperfusion.
[Hold] (ShieldAlert Icon)	Table Action Btn	Issues instant emergency regulatory quarantine on the organ, locking all downstream transfers on-chain.

4.3 Organ Provenance Inspector (VerifyPage) Actions

Control Element	Type	Action & Operational Effect
Organ Registry ID Input	JTextField	Accepts specimen barcode string (e.g. ORG-KID-8041, ORG-HRT-2094).
[Interrogate Ledger]	JButton Submit	Dispatches cryptographic REST query to /api/verify/:organId and renders verified provenance timeline.
[Scan QR / RFID]	JButton Tool	Opens QRScannerModal to scan physical packaging QR using device camera or file upload.
Preset: ORG-KID-8041	JButton Preset	Instantly loads kidney specimen in active cryo-transit (O+ blood group, 24h limit, Custodiol perfusion).
Preset: ORG-HRT-2094	JButton Preset	Instantly loads critical heart specimen in air transit under 4-hour emergency ischemia timer.
Preset: ORG-LVR-5512	JButton Preset	Instantly loads liver specimen received in surgical suite ready for operating theater implantation.
Preset: ORG-COR-3390	JButton Preset	Demonstrates regulatory quarantine lock state due to biosafety or chain-of-custody discrepancy.
Preset: ORG-FAKE-999	JButton Preset	Demonstrates anti-trafficking counterfeit detection (returns 404 UNREGISTERED ORGAN alert).
[Donor Consent (Form 8)]	JButton Dialog	Displays printable Form 8 brain-death clearance certificate anchored to IPFS multihash.
[Flag Incident]	JButton Danger	Opens ReportModal to file an emergency whistleblower trafficking or temperature breach alert.
[Copy TxHash]	JButton Action	Copies the cryptographic transaction hash of any custody milestone directly to clipboard.

4.4 Modal Dialogs (JDialogs) & Action Buttons

Modal Dialog (JDialog)	Key Buttons / Controls	Action & Operational Effect
Harvest Organ Modal (CreateBatchModal)	• Template Presets (Kidney/Heart/Liver) • [OK - Sign & Register Harvest] • [Cancel] / [X]	Populates donor genetic data. Executes POST /api/organs creating immutable EVM block record and sets cold-ischemia countdown.
Custody Transfer Modal (TransferModal)	• Consortium Entity Dropdown • Milestone Location Field • [OK - Sign Medical Handoff]	Selects recipient stakeholder node. Signs custody transfer on-chain with real-time temperature telemetry (2.4°C).
Specimen Manifest Modal (QRCodeModal)	• [Copy Verify URL] • [Print Carrier Tag] • [X] Close	Copies browser verification URL or triggers window.print() for physical cryo-shipper box RFID/QR barcode label.
Consent Certificate Modal (CertificateModal)	• [Print Form 8 Certificate] • [X] Close	Formats official legal Form 8 Deceased Donor Certificate with IPFS multihash stamp and surgeon signature for printing.
Trafficking Alert Modal (ReportModal)	• Violation Category Dropdown • [OK - Dispatch Regulatory Alert] • [Switch to Regulatory Admin]	Files immediate whistleblower report to NOTTO auditors and allows instant switching to Admin workstation to quarantine the organ.
Optical QR Scanner (QRScannerModal)	• [Webcam Scanner] Tab • [Demo Presets] Tab • [Load QR Image File]	Decodes QR barcodes via Html5Qrcode camera stream or uploaded graphics files, automatically launching verification.
Defense & Viva Guide (PresentationGuideModal)	• [Live Demo Flow] Tab • [Viva Defense] Tab • [Smart Contract Rules] Tab	Guides student / presenter through 1-minute chronological demonstration sequence and examiner viva arguments.
Bottom Status Strip (JStatusBar)	• [API Health] Link • [REST JSON] Link • [Defense Guide [F1]]	Provides one-click diagnostics: tests port 5000 server health, inspects raw JSON data for ORG-KID-8041, and launches guide.

5. Step-by-Step Operator & Demonstrator Workflow

To perform a flawless live demonstration or operational audit, follow this chronological sequence:

Step 1: OPO Node Harvest & Registration

1. In the top JToolBar, switch the consortium actor dropdown to 'Organ Procurement Org (OPO)'.

2. Click the beveled '[Harvest Organ [F2]]' button to launch the JDialog registration modal.

3. Select a preset template (e.g. 'Kidney (24h)' or 'Heart (4h Urgent)'). Notice automatic Form 8 donor consent hash generation.

4. Click '[OK - Sign & Register Harvest]'. The organ is immediately anchored on the EVM blockchain.

Step 2: Cryo-Logistics Transit & Telemetry Logging

1. Switch the actor dropdown to 'Medical Transit Logistics' (AeroMed Logistics).

2. In the Organ Custody Registry, find your harvested organ and click the '[Transfer]' button.

3. Select the destination node (NOTTO Clearinghouse or Hospital) and record probe temperature (2.4°C).

4. Click '[OK - Sign Medical Handoff]'. Stage transitions to 'InTransit'.

Step 3: National Allocation Clearance & Recipient Matching

1. Switch the actor dropdown to 'Allocation Authority' (NOTTO Regional Directorate).

2. Audit HLA histocompatibility (6/6 match) and verify recipient queue priority.

3. Execute handoff to the designated surgical hospital. Stage transitions to 'AllocationCleared'.

Step 4: Hospital Operating Suite Reception & Implantation

1. Switch the actor dropdown to 'Transplant Hospital' (Apollo Multi-Specialty Surgical Center).

2. Notice the organ is now in 'HospitalReceived' status in the surgical suite.

3. Click the '[Implant]' button. Enter surgical operating theater notes.

4. Implantation is completed on-chain. State transitions permanently to 'Transplanted' (Grafted).

Step 5: Public Provenance Inspection & Anti-Trafficking Audit

1. Navigate to 'Provenance' (VerifyPage).

2. Click '[ORG-KID-8041]' or enter an organ ID to interrogate the ledger.

3. Inspect the live cold ischemia window countdown, donor Form 8 IPFS CID, and complete custody timeline.

4. Click '[ORG-FAKE-999]' to illustrate automated detection of unregistered organs and illegal trafficking alerts.

6. Academic Defense: Why Blockchain vs. Traditional MongoDB/SQL?

When defending this project before examiners or regulatory auditors, emphasize these four foundational arguments:

1. Multi-Stakeholder Mutual Distrust: OPOs, private air-ambulances, rival private hospital corporations, and federal health ministries do not trust each other with database administrative access. A centralized MongoDB database managed by Hospital A could allow a rogue DBA to covertly alter timestamps, swap recipient priorities, or delete cold-chain failure alerts.

2. Anti-Trafficking Non-Repudiation: Under human organ transplant laws (NOTA/THOTA), black-market commercial trafficking relies on forged donor consent. On LifeLedger, every organ requires cryptographic digital signatures tied to the OPO's private key and Form 8 IPFS CID. Once committed to an EVM block, provenance is immutable and permanent.

3. Programmable Invariant Enforcement: Smart contracts execute programmable law, not passive storage. If a cold-ischemia window has expired, or if a non-hospital entity attempts to sign off on a transplant, the EVM reverts the transaction with mathematical finality.

4. Zero-Volatility Proof-of-Authority Consensus: LifeLedger uses a PoA consortium consensus model where authenticated health ministry entities validate blocks. This eliminates volatile public gas fees and guarantees predictable 1-2 second transaction finality during medical emergencies.